

# Honors Physics - Spring Final Exam Review

## Multiple Choice

### Ch 8: Momentum

#### Vocab

- momentum
- impulse
- inelastic collision
- conservation of momentum
- elastic collision

#### Questions

1. What is conservation of momentum?
  - A. the total momentum before a collision is equal to the total momentum after a collision
  - B. impulse equals change in momentum
  - C. two objects stick together after a collision
  - D. total momentum of a system is always zero
2. Why does a gun recoil when a bullet is shot from it?
  - A. the forward motion of the bullet imparts an impulse on the gun that pushes it backwards
  - B. the force of the explosion causes a push on the gun
  - C. the forward momentum of the bullet needs to be counteracted by a backward momentum in the gun
3. Which of the following effects would not decrease the force on an object (such as a dropped egg)?
  - A. decrease the velocity of the object before the collision
  - B. decrease the time of collision
  - C. increase the time of collision
  - D. decrease the velocity of the object after the collision
4. After a car accident, the two cars stick together. This is an example of a/an \_\_\_\_
  - A. inelastic collision
  - B. impulse collision
  - C. elastic collision
  - D. destructive collision
5. How can you use conservation of momentum to explain why a water rocket flies
  - A. objects on Earth cannot zero momentum, so they must move
  - B. if water is forced out the bottom of the rocket, the rocket must launch upwards to conserve momentum
  - C. putting water in the rocket gives it an initial momentum in the upward direction
  - D. the interaction between the water and the rocket is an inelastic collision, giving it momentum
6. A bouncy ball and a sphere of putty—each of the same mass—are thrown at a wall with the same initial velocity. The putty sticks to the wall while the ball rebounds back in the direction it came with almost the same velocity. Which of the following is true?
  - A. The bouncy ball has a greater change in momentum
  - B. The putty has a greater change in momentum
  - C. The ball and the putty have the same change in momentum
  - D. None of these are true.

7. Consider two unequal masses,  $M$  and  $m$ . Which of the following statements is false?
- A. The center of mass is closer to the larger mass.
  - B. The center of mass lies on the line joining the centers of each mass.
  - C. There is only one center of mass
  - D. It is possible for the center of mass to lie within one of the objects.
  - E. none of the above
8. When a cannon fires a cannonball, the cannon will recoil backward because the
- A. energy of the cannon is greater than the energy of the cannonball.
  - B. momentum of the cannonball and cannon is conserved.
  - C. momentum of the cannon is greater than the energy of the cannonball.
  - D. energy of the cannonball and cannon is conserved.
9. A small car meshes with a large truck in a head-on collision. Which of the following statements concerning the magnitude of the average collision force is correct?
- A. The small car and the truck experience the same average force.
  - B. The small car experiences the greater average force.
  - C. The truck experiences the greater average force.
  - D. It is impossible to tell since the masses and velocities are not given.
10. In an inelastic collision, if the momentum is conserved, then which of the following statements is true about kinetic energy?
- A. Kinetic energy is gained.
  - B. Kinetic energy is also conserved.
  - C. Kinetic energy is lost.
  - D. none of the above
11. A car of mass 2000 kg moves to the right along a level, straight road at a speed of 6.0 m/s. It collides directly with a stopped motorcycle of mass 100 kg. What is the total momentum after the collision?
- A. 2000 kg m/s to the right
  - B. 6000 kg m/s to the right
  - C. 12.000 kg m/s to the right
  - D. zero
  - E. none of the above
12. A 2000-kg car traveling at 20 m/s runs into the rear of a stopped car that has a mass of 500 kg and they stick together. What is the speed of the cars after the collision?
- A. 10 m/s   B. 5.0 m/s   C. 16 m/s   D. 20 m/s   E. none of the above

## Ch 10: Rotation

### Vocab

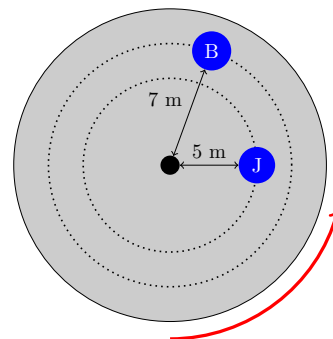
- angular velocity
- torque
- angular momentum
- angular acceleration
- radians
- moment of inertia

### Questions

- An ice skater performs a pirouette (a fast spin) by pulling in his outstretched arms close to his body. What happens to his **angular momentum** about the axis of rotation?  
A. it increases.    B. it decreases.    C. it stays the same    D. impossible to say.
- An ice skater performs a pirouette (a fast spin) by pulling in his outstretched arms close to his body. What happens to his **moment of inertia** about the axis of rotation?  
A. it increases.    B. it decreases.    C. it stays the same    D. impossible to say.
- Angular velocity is expressed in units of  
A. radians per second.    B. meters per second.    C. arcs per second.    D. omegas per second.
- Angular acceleration is expressed in units of  
A. arcs per second squared.    C. alphas per second squared.  
B. meters per second squared.    D. radians per second squared.
- Two equal forces are applied to a door. The first force is applied at the midpoint of the door; the second force is applied at the doorknob. Both forces are applied perpendicular to the door. Which force exerts the greater torque?  
A. the second at the doorknob    C. both exert zero torques  
B. the first at the midpoint    D. both exert equal non-zero torques
- Two forces are applied to a doorknob, perpendicular to the door. The first force is three times as large as the second force. The ratio of the torque of the first to the torque of the second is  
A. 1/4.    B. 2.    C. 1/3.    D. 3.
- If a constant net torque is applied to an object, that object will  
A. have a decreasing moment of inertia.  
B. rotate with constant angular displacement.  
C. rotate with constant angular velocity.  
D. having an increasing angular velocity.
- Two spheres have the same radius and equal masses. One is made of solid aluminum, and the other is made from a hollow shell of gold. If allowed to race down an inclined plane, which one will reach the bottom first?  
A. the solid aluminum sphere will reach the bottom first  
B. the hollow gold sphere will reach the bottom first  
C. they will reach the bottom at the same time  
D. it is impossible to tell
- What is the quantity used to measure an object's resistance to changes in rotation?  
A. angular velocity ( $\omega$ )    B. moment of inertia ( $I$ )    C. mass ( $m$ )    D. torque ( $\tau$ )

22. Little Bobby and Janie are both riding a carousel. Janie is on a pony, which is 5 m from the center of rotation, and Bobby is on a water buffalo, which is 7 m from the center. The merry-go-round makes one complete revolution every 30 seconds. Which of the following is true?

- A. Bobby has larger  $v$ ; Janie has larger  $\omega$
- B. Bobby has larger  $\omega$ ; Janie has larger  $v$
- C. Bobby has larger  $v$ ; both kids have the same  $\omega$
- D. Bobby has larger  $\omega$ ; both kids have the same  $v$
- E. Both kids have the same  $\omega$  and the same  $v$



## Ch 16: SHM & Waves

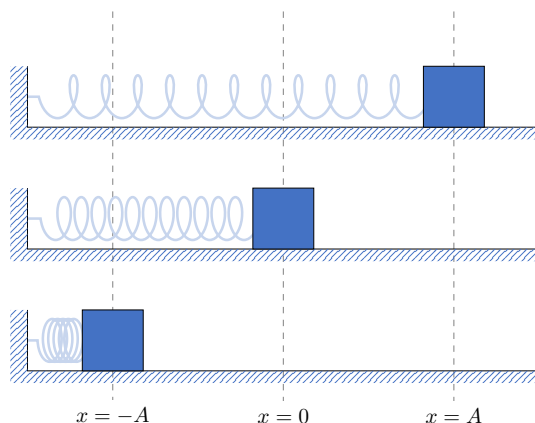
### Vocab

- simple harmonic motion
- frequency
- period
- spring constant
- amplitude
- interference
- longitudinal wave
- transverse wave
- resonance

### Questions

23. What would happen to the period of a pendulum if you increase the mass?  
 A. period would increase    B. period would decrease    C. period wouldn't change
24. What would happen to the period of a *spring* if you increase the mass?  
 A. period would increase    B. period would decrease    C. period wouldn't change
25. For a spring, what causes the restoring force?  
 A. inertia of the mass    C. tension of the spring itself  
 B. force of gravity    D. there is no restoring force
26. Increasing the frequency \_\_\_\_\_ the period.  
 A. does not affect    B. decreases    C. increases
27. Why does a pendulum keep moving when it gets to equilibrium?  
 A. the potential energy of the pendulum pulls it up from equilibrium back to the top  
 B. the inertia of the pendulum makes it want to keep moving through equilibrium  
 C. the restoring force pushes it past equilibrium  
 D. the frequency of the pendulum keeps it from stopping
28. A wave in which the particles back-and-forth parallel to the direction of the wave's motion is known as a \_\_\_\_\_.  
 A. simple harmonic wave    B. longitudinal wave    C. transverse wave    D. destructive interference

A mass is attached to a spring sitting on a frictionless horizontal surface. The mass oscillates between the positions  $x = -A$  and  $x = A$  as shown:



29. Indicate where the mass will have a velocity of zero.  
A. at  $x = \pm A$       B. at  $x = 0$       C. at  $x = \pm A/2$       D. never
30. Indicate where the mass will have an acceleration of zero.  
A. at  $x = \pm A$       B. at  $x = 0$       C. at  $x = \pm A/2$       D. never
31. What do waves do?  
A. transport work without transporting power  
B. transport energy without transporting matter  
C. move particles from one place to another without transporting energy  
D. absorb forces while generating energy
32. In a wave, the maximum displacement of points of the wave from equilibrium is called the wave's  
A. frequency    B. speed    C. trough    D. wavelength    E. none of the above
33. Microwaves have a higher frequency than radio waves. Therefore, which of the following is true?  
A. microwaves have a longer wavelength than radio waves  
B. microwaves have a shorter wavelength than radio waves  
C. microwaves have a faster speed than radio waves  
D. microwaves have a slower speed than radio waves
34. Two simple pendulums A and B are 3.0 m long. The period of A is  $T$ . Pendulum A is three times as heavy as pendulum B. What is the period of B?  
A.  $T$       B.  $0.71T$       C.  $2T$       D.  $1.4T$
35. How can sound be used to break glass?  
A. if sound is played at a sufficiently high frequency the glass will resonate  
B. if sound is played at a certain frequency the glass will resonate  
C. if sound is played at a sufficiently high volume the glass will resonate  
D. if sound is played at a sufficiently high wavelength the glass will resonate

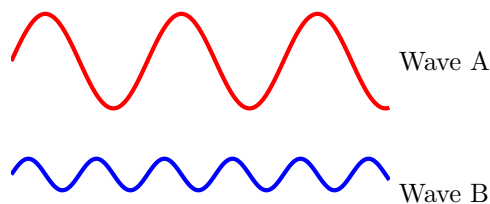
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36. The distance between successive troughs on a wave is called the wave's  
A. frequency                      B. speed                      C. wavelength                      D. amplitude
37. Resonance in a system, such as a string fixed at both ends, occurs when  
A. its frequency is smaller than the frequency of an external source.  
B. it is oscillating in simple harmonic motion.  
C. its frequency is greater than the frequency of an external source.  
D. its frequency is the same as the frequency of an external source.
38. The time for one cycle of a periodic process is called the  
A. amplitude.                      B. period.                      C. wavelength.                      D. frequency.
39. A 1.0-kg object is attached to a spring of spring constant 10 N/m. The object is displaced by 5.0 cm from the equilibrium position and let go. What is the period of vibration?  
A. 8.0 s                      B. 4.0 s                      C. 16 s                      D. 2.0 s
40. For a periodic process, the number of cycles per unit time is called the  
A. period.                      B. amplitude.                      C. wavelength.                      D. frequency.
41. Consider the two waves below. Assume that they are both traveling through the same medium. Which of the following is true?



- A. Wave B has both a larger wavelength and a higher frequency  
B. Wave B has a larger wavelength, but Wave A has a higher frequency  
C. Wave A has both a larger wavelength and a higher frequency  
D. Wave A has a larger wavelength, but Wave B has a higher frequency

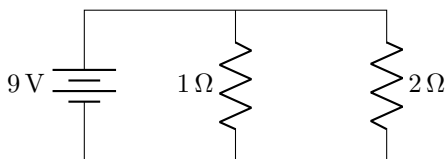
**Ch 18-21: Electricity****Vocab**

- Conductors
- Potential difference/Voltage
- Ohm's Law
- Electric Field
- Ground
- Circuits
- Current
- Resistance
- Test charges

**Questions**

42. Two lightbulbs are wired in a series circuit. What happens when one of the bulbs breaks?
- A. the other bulb stays just as bright
  - B. both bulbs go out
  - C. the other bulb gets dimmer
  - D. the other bulb gets brighter
43. Two lightbulbs are wired in a parallel circuit. What happens when one of the bulbs breaks?
- A. the other bulb stays just as bright
  - B. both bulbs go out
  - C. the other bulb gets dimmer
  - D. the other bulb gets brighter
44. Which of the statements is true?
- A. resistance and current are directly proportional
  - B. voltage and current are directly proportional
  - C. resistance and charge are directly proportional
  - D. two of the above are true
45. Which of the following is NOT true for a series circuit?
- A. The voltage across each resistor is the same as the battery's voltage
  - B. The current through each resistor is the same as the current out of the battery
  - C. The sum of the voltages across each resistor is the same as the battery's voltage
  - D. The equivalent resistance is larger than the resistance of any individual resistor
46. Which of the following is NOT true for a parallel circuit?
- A. The voltage across each resistor is the same as the battery's voltage
  - B. The current through each resistor is the same as the current out of the battery
  - C. The sum of the current through each resistor is the same as the current out of the battery
  - D. The equivalent resistance is smaller than the resistance of any individual resistor
47. An ampere is a unit of electrical
- A. voltage
  - B. resistance
  - C. pressure
  - D. all of these
  - E. none of these
48. Two protons attract each other gravitationally and repel each other electrically. By far the smaller is
- A. the electrical repulsion.
  - B. the gravitational attraction.
  - C. neither - they are the same.
49. Two charged particles attract each other with a force  $F$ . If the charges of both particles are tripled, and the distance between them tripled, then the force of attraction will be
- A.  $F/4$ .
  - B.  $F/2$ .
  - C.  $F$ .
  - D.  $2F$ .
  - E. none of these.
50. Two charged particles repel each other with a force  $F$ . If the charge of one of the particles is doubled and the distance between them is doubled, then the force will be
- A.  $F/2$ .
  - B.  $2F$ .
  - C.  $F/4$ .
  - D.  $F$ .
  - E. none of these.

51. Two charged objects are separated by a distance  $d$ . The first charge is larger in magnitude than the second charge.
- A. The charges exert forces on each other equal in magnitude and pointing in the same direction.
  - B. The second charge exerts a larger force on the first charge.
  - C. The first charge exerts a larger force on the second charge.
  - D. The charges exert forces on each other equal in magnitude and opposite in direction.
52. Two charges separated by one meter exert 1-N forces on each other. If the charges are pushed to  $1/2$  meter separation, the force on each charge will be
- A. 1 N.                      B. 2 N.                      C. 4 N.                      D. 8 N.                      E. 16 N.
53. The total amount of charge that passes through a wire's full cross section at any point per unit of time is referred to as a
- A. voltage.    B. current.    C. wattage.    D. electric potential.
54. Consider the circuit below. Which of the following is true?



- A. the 1-ohm resistors has more current flowing through it than the 2-ohm resistor
  - B. the 2-ohm resistors has more current flowing through it than the 1-ohm resistor
  - C. there is a larger potential difference accross the 1-ohm resistor than the 2-ohm resistor
  - D. there is a larger potential difference accross the 2-ohm resistor than the 1-ohm resistor
  - E. Both (a) & (c)
  - F. Both (b) & (d)
55. What happens to the current through the battery when more resistors are added in series.
- A. current increases    B. current decreases    C. current stays the same    D. impossible to say
56. What happens to the current through the battery when more resistors are added in parallel?
- A. current increases    B. current decreases    C. current stays the same    D. impossible to say
57. A main difference between gravitational and electric forces is that electrical forces
- A. attract.
  - B. are weaker.
  - C. act over shorter distances.
  - D. obey the inverse-square law.
  - E. are much stronger.



## Ch 25: Geometric Optics

### Vocab

- refraction
- reflection
- real image
- virtual image
- focal length

### Questions

58. The law of reflection states:

- A. incoming light reflects along the same path that it came
- B. light bends when it goes from one medium to another
- C. you can only see your reflection on smooth surfaces
- D. the angle of incidence is equal to the angle of reflection

59. What is refraction?

- A. when light bends because it changes speed
- B. when light bends because it changes frequency
- C. when light changes color after going through a prism
- D. when light bounces back through the medium it came from

60. How does a lens work?

- A. the lens disperses light toward the focal point
- B. the lens interferes light toward a focal point
- C. the lens reflects light toward a focal point
- D. the lens refracts light toward a focal point

61. The angle of reflection

- A. is always less than the angle of incidence.
- B. is always greater than the angle of incidence.
- C. may be greater than or less than the angle of incidence.
- D. none of the above

62. The principle on which mirrors work is

- A. polarization.
- B. dispersion.
- C. refraction.
- D. reflection.

63. Light arriving at a concave mirror on a path through the center is reflected

- A. through the focal point.
- B. back on itself.
- C. back parallel to the axis.
- D. through the vertex of the mirror.

64. Which of the following is true about light when it enters a new medium that causes its speed to slow down.

- A. the wavelength of the light increases
- B. the wavelength of the light decreases
- C. the frequency of the light increases
- D. the frequency of the light increases

65. In a converging lens, where must the object be in order for the image to be real?
- A. inside the focal point
  - B. outside the focal point
  - C. at the focal point
  - D. converging lenses never make real images
66. A spherical mirror on which reflection takes place on the inner surface of the sphere is referred to as a
- A. convex mirror.
  - B. concave mirror.
67. A spherical mirror on which reflection takes place on the outer surface of the sphere is referred to as a
- A. convex mirror.
  - B. concave mirror.

## Problems

68. A 1550-kg car moving south at 10 m/s collides with a 2550-kg car moving north. The cars stick together and move as a unit after the collision with a velocity of 5.22 m/s to the north. Find the velocity of the 2550-kg car before the collision.
69. James Bond is sitting in the middle of a frictionless pond and left to die by an evil villain who knows nothing about physics. Bond, who is at rest and has a mass of 75.2 kg (including his shoes) takes one of his shoes ( $m = 2.2$  kg) and throws it in the opposite direction he wants to move with a velocity of 20 m/s. With what velocity will Bond move?
70. The spring in a grocery produce scale is stretched 0.03 meters when 2 kg of oranges are placed in the basket. (a) What is the spring constant? (b) If allowed to oscillate, what will the period of oscillation be?
71. A pendulum makes 9 oscillations in 15 seconds. Find the period and frequency of the pendulum.

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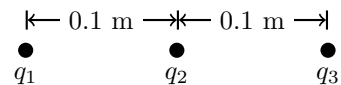
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72. Red light has a wavelength of  $5.90 \times 10^{-7}$  m. What is the frequency of red light? The speed of light is  $3.00 \times 10^8$  m/s.

73. A charge  $Q$  of  $+7.7 \times 10^{-5}$  C sits at the origin.  
(a) What is the electric field at  $x = +0.64$  meters? (b) If a charge of  $-1.8 \times 10^{-6}$  C is placed at that location, what force will it experience? (make sure to include directions!)

74. Three charges ( $q_1 = 4.1 \times 10^{-5}$  C,  $q_2 = 6.6 \times 10^{-5}$  C, and  $q_3 = -1.3 \times 10^{-5}$  C) are situated 0.1 meters from each other. What is the force acting on  $q_2$ ?



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Period:

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75. A 200-ohm resistor is connected across 220 V. What current will flow?
76. What potential difference is required to cause 4.00 A to flow through a resistance of 120 ohms?
77. Consider a **diverging** lens with a focal length of 3 cm. An object is 4 cm away. (a) What is the distance between the lens and the image? (b) What is the magnification of the image? (c) Is the image real or virtual? How do you know?
78. A beam strikes a pane of glass at an angle of 32 degrees from the normal. If the index of refraction of the glass is 1.56, and the index of refraction of air is approximately 1.00, what is the angle of refraction?

## Answers

**Ch 8: Momentum**

1. A
2. C
3. B
4. A
5. B
6. A
7. E
8. B
9. A
10. C
11. C
12. C

**Ch 10: Rotation**

13. C
14. B
15. A
16. D
17. A
18. D
19. D
20. A
21. B
22. C

**Ch 16: SHM & Waves**

23. C
24. A
25. C
26. B

27. B
28. B
29. A
30. B
31. B
32. E
33. B
34. A
35. B
36. C
37. D
38. B
39. D
40. D
41. D

**Ch 18-21: Electricity**

42. B
43. A
44. B
45. A
46. B
47. E
48. B
49. C
50. A
51. D
52. C
53. B
54. A

55. B
56. A
57. E

**Ch 25: Geometric Optics**

58. D
59. A
60. D
61. D
62. D
63. B
64. B
65. B
66. B
67. A

**Problems**

68. 14.47 m/s
69. 0.603 m/s
70. (a) 653.3 N/m; (b) 0.348 s
71.  $T = 1.67$  s;  $f = 0.6$  Hz
72.  $5.08 \times 10^{14}$  Hz
73. (a)  $1.69 \times 10^6$  N/C (+);  
(b) 3.05 N (-)
74. 3207.6 N
75. 1.1 A
76. 480 V
77. (a) -1.71 cm; (b) 0.43; (c)  
virtual because  $d_i < 0$
78.  $19.9^\circ$